Symmetries and Lift-and-Project Hierarchies

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Motivation and Goals

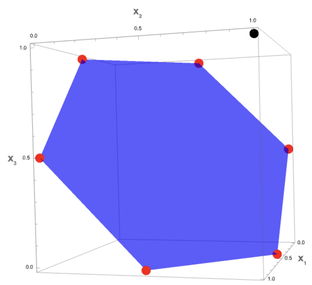
- At least 36% of MIPLIB 2010 problems have nontrivial symmetries.
- It is folklore knowledge that symmetry "harms" IP solvers.

Goals

- Investigate how symmetry harms IP solvers, in particular for improving the LP bounds, and
- Find bounds that depend only on the group-theoretical properties, independent of the specific class of polytopes.

Symmetric IPs

$$\begin{aligned} \min \quad & x_1 + x_2 + x_3 \\ & x_1 - 2x_2 - 2x_3 \geq -3 \\ & -2x_1 + x_2 - 2x_3 \geq -3 \\ & -2x_1 - 2x_2 + x_3 \geq -3 \\ & 2x_1 + 2x_2 + 2x_3 \geq 3 \\ & x \in \{0,1\}^3 \end{aligned}$$



- Suppose $(\overline{x}_1, \overline{x}_2, \overline{x}_3)$ is feasible for the LP relaxation.
- Is $(\overline{x}_2, \overline{x}_1, \overline{x}_3)$ feasible? Yes? Any **permutation** of $(\overline{x}_1, \overline{x}_2, \overline{x}_3)$ is feasible.

Symmetries

Feasible permutations of $x = (x_1, \dots, x_n)$ for every feasible x.

k-transitivity

$P \subseteq [0,1]^n$ given by a description of inequalities $Ax \geq b$.

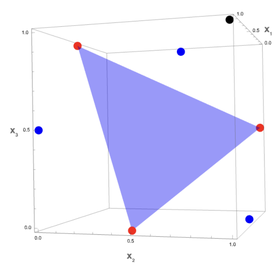
P is **k-transitive**: $\forall x \in P$, and $\forall$ pair of ordered tuples $(i_1, \dots, i_k)$ and $(j_1, \dots, j_k)$ of $[n]$, there exist $z \in P$ permutation of x, where the entry at position i_ℓ of x is mapped to position j_ℓ of z.

$$x = \begin{pmatrix} \vdots \\ x_{i_1} \\ x_{i_2} \\ \vdots \\ x_{i_{k-1}} \\ x_{i_k} \\ \vdots \end{pmatrix} \rightsquigarrow \begin{pmatrix} x_{i_2} \\ \vdots \\ x_{i_k} \\ \vdots \\ x_{i_{k-1}} \\ \vdots \\ x_{i_1} \end{pmatrix} = \begin{pmatrix} z_{j_1} \\ \vdots \\ z_{j_2} \\ \vdots \\ z_{j_{k-1}} \\ \vdots \\ z_{j_k} \end{pmatrix} = z$$

STS is 2-transitive

$$\begin{array}{cccccccccc} x_1 & +x_2 & +x_3 & & & & & & & \geq 1 \\ & & & x_4 & +x_5 & +x_6 & & & & \geq 1 \\ & & & & & & x_7 & +x_8 & +x_9 & \geq 1 \\ x_1 & & & +x_4 & & & +x_7 & & & \geq 1 \\ & x_2 & & & +x_5 & & & +x_8 & & \geq 1 \\ & & x_3 & & & +x_6 & & & +x_9 & \geq 1 \\ x_1 & & & & +x_5 & & & & +x_9 & \geq 1 \\ x_1 & & & & & +x_6 & & +x_8 & & \geq 1 \\ & x_2 & & +x_4 & & & & & +x_9 & \geq 1 \\ & x_2 & & & & +x_6 & +x_7 & & & \geq 1 \\ & & x_3 & +x_4 & & & & +x_8 & & \geq 1 \\ & & x_3 & & +x_5 & & +x_7 & & & \geq 1 \end{array}$$

P below is 1-transitive



$P := \text{conv}(\{(1, 1/2, 0), (0, 1, 1/2), (1/2, 0, 1)\})$

Lift-and-Project Hierarchies

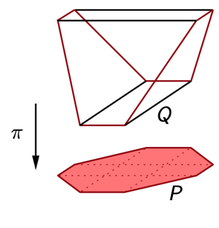
A polytope $Q \subseteq [0,1]^{n+r}$ is an extended formulation of P if P is a linear projection of Q.

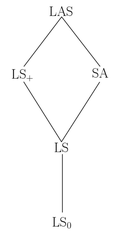
L&P Operator

$T : \mathbb{R}^n \to \mathbb{R}^{n+r}$

- $T(P)$ is convex, $\pi_x(T(P)) \subseteq P$;
- $\pi_x(T(P)) \cap \{0,1\}^n = P \cap \{0,1\}^n$;
- $\forall S, R \subseteq [0,1]^n$ convex, $S \subseteq R \implies T(S) \subseteq T(R)$.

$T^k(\cdot)$: operator T applied k times





Our Main Result

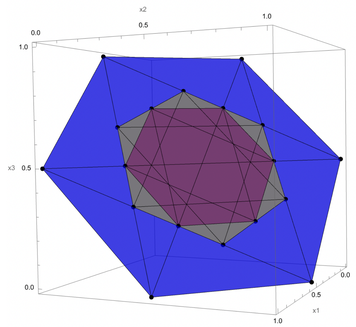
Theorem 1

Let $P \subseteq [0,1]^n$ be a $(k+1)$-transitive polytope with $\dim P \leq n-1$ and without integer points. The following are equivalent:

1) $\text{LS}_0^k(P) \neq \emptyset$
2) $\text{SA}^k(P) \neq \emptyset$
3) $\text{LS}_+^k(P) \neq \emptyset$
4) P intersects all of the $(n-k)$-dimensional faces of $[0,1]^n$.

Key Takeaways

- The more symmetric the optimal LP face, the harder to cut!
- "Hierarchy collapse" of L&P operators under symmetry



Theorem 1'

Let $P \subseteq [0,1]^n$ be a $(k+1)$-transitive with $\dim P \leq n-1$ and no integer points. P intersects all the $(n-k)$-dimensional faces of $[0,1]^n$ iff there exists $\overline{x} \in \text{SA}^k(P)$ with a lifting $\hat{x} \in [0,1]^{2^n}$ given by

$$\hat{x}_S = \begin{cases} \frac{(\|x\|_1)_{|S|}}{(n)_{|S|}} & \text{if } |S| \leq k+1 \\ 0 & \text{o.w.} \end{cases}$$

An Example – Knapsack

For $n \in 2\mathbb{N}$,

$$K_n := \left\{ x \in [0,1]^n \, : \, \sum_{i=1}^n x_i = (n+1)/2 \right\}.$$

A result by Grigoriev (2001) implies that Sherali-Adams level $k \in \Theta(n)$ is needed to certify that K_n is integer empty, i.e., $\text{SA}^k(K_n) = \emptyset$.

Extensions and Work in Progress

Theorem 2

Let $P \subseteq [0,1]^n$ be a 2-transitive polytope such that $\mathbf{0}, \mathbf{1} \notin P$. The following are equivalent:

1. $\text{LS}_0(P) \neq \emptyset$;
2. $\text{LS}(P) \neq \emptyset$;
3. P intersects all of the facets of $[0,1]^n$.

This result is tight with respect to the assumption on transitivity.

- Work in progress: Extension of Theorem 2 to an arbitrary number of rounds.
- Group action on a subset of variables.

SA and LS_0

1 $\forall i \in [n]$, do $x_i \cdot (Ax - b \geq 0)$ and $(1 - x_i) \cdot (Ax - b) \geq 0$

2 Linearize $x_i x_j$ as y_{ij} and set $x_i^2 = x_i$

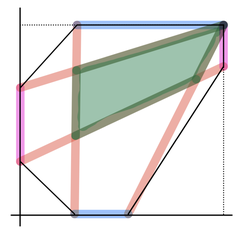
$$M(P) := \left\{ (x,y) \in \mathbb{R}^n \times \mathbb{R}^{\binom{n}{2}} \, : \, \text{linearized inequalities} \right\}$$

3 $\text{SA}(P) := \{x \in \mathbb{R}^n \, : \, x \in \text{proj}_x(M(P))\}$

Decomposition Lemma (one round)

Let $(x,y) \in [0,1]^n \times [0,1]^{\binom{n}{2}}$. Then $(x,y) \in M(P)$ iff $\forall i \in [n]$,

- if $x_i \in (0,1)$, then $\exists \overline{z}^i, \underline{z}^i \in P$ s.t. $\forall j \in [n]$, $\quad \overline{z}^i_j = \frac{y_{ij}}{x_i} \quad \underline{z}^i_j = \frac{x_j - y_{ij}}{1 - x_i}$,
- if $x_i \in \{0,1\}$, then $\exists z^i \in P$ s.t. $\forall j \in [n]$, $\quad z^i_j = x_j$.



Barycenter Lemma

Let G be the group of permutations of P. If G is transitive (P is 1-transitive), then

$$\text{bary}_G(x) := \frac{1}{|G|} \sum_{g \in G} gx = \gamma_x \mathbf{1}$$

for every $x \in P$, where $\gamma_x \in [0,1]$. Note that $\text{bary}_G(x) \in P$ since P is convex.

Proof Sketch Theorem 2: 3 $\implies$ 2

- By assumption the polytopes $\overline{P} := P \cap \{x : x_1 = 1\}$ and $\underline{P} := P \cap \{x : x_1 = 0\}$ are nonempty.
- Hence there exist $\overline{x} := (1, \overline{x}_2, \dots, \overline{x}_n) \in P$ and $\underline{x} := (0, \underline{x}_2, \dots, \underline{x}_n) \in P$.
- **2-transitivity**: $\exists$ a subgroup of permutation symmetries (for $\overline{P}$ and $\underline{P}$) that fixes 1 and is transitive on $\{2, \dots, n\}$.
- **Average on $\{2, \dots, n\}$**: There exist $\overline{z}^1 := (1, \alpha, \dots, \alpha) \in P$ and $\underline{z}^1 := (0, \beta, \dots, \beta) \in P$, with $\alpha, \beta \in (0,1)$.
- Given α, β it is easy show that there exists $\gamma \in (0,1)$ such that $\gamma \mathbf{1} = \gamma \overline{z}^1 + (1-\gamma)\underline{z}^1$ which implies $\beta = \frac{1-\alpha\gamma}{1-\gamma}$.
- **Exhibit a lifted point**: Define $(x,y) \in [0,1]^n \times [0,1]^{\binom{n}{2}}$ as $x_i := \gamma$ for all i and $y_{ij} := \alpha\gamma$ for all $i \neq j$.
- **Check conditions**: for $j \neq 1$, $\frac{y_{ij}}{x_i} = \alpha = \overline{z}^1_j$ and $\frac{1-y_{ij}}{1-x_i} = \frac{1-\alpha\gamma}{1-\gamma} = \beta = \underline{z}^1_j$; $\quad$ for $j = 1$, $\frac{x_i}{x_i} = 1 = \overline{z}^1_1$ and $\frac{x_i - x_1}{1-x_1} = 0 = \underline{z}^1_1$.
- **Decomposition Lemma**: Permutations of $\overline{z}^1, \underline{z}^1$ give us $\overline{z}^i, \underline{z}^i \in P$ for $i \in [n]$ for which the same (x,y) works.

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